

QUESTION-TO-SLIDE MAPPING

SECTION A - LECTURE 04

Climate & Built Environment

Course	BECM 3115 - Climate and Architectural Design
Lecture deck	4.0 - Climate & Build Environment (8 PDF pages/slides)
Question bank	Regular term-final examinations, 2016-2024 (8 supplied papers; 2019 not present)
Method	Topic-level matching to the eight-slide Lecture 04 deck. Because this lecture explicitly hands off from the "accept/reject/mitigate" problem to designer responsibilities, essential knowledge and complexities, those blocks are treated as the authoritative scope. Questions requiring complete passive/active thermal-control comparison are marked Partial because the deck only names passive and active design technology as knowledge domains.

RESULT

Lecture 04 is a compact but high-value responsibility/knowledge lecture. The strongest matches are the repeated built-form designer responsibility and complexity questions (2017, 2020, 2021, 2022, 2024), the 2017/2024 essential-knowledge questions, and the 2018 "accept fate / reject development / mitigate" question. The deck is short, so most of its eight slides carry direct exam value.

Prepared as part of the cumulative lecture-by-lecture question bank analysis.

1. Executive Summary

Lecture 04 answers the question raised at the end of Lecture 02: what should the built-form designer do now? It frames climate-responsive design as mitigation rather than rejection of development, then specifies external/internal responsibilities, six essential knowledge domains, and six major design complexities. Historical papers repeatedly test these ideas, sometimes under thermal comfort and sometimes under broad climate-design wording.

8	10	5	1	4	130
Slides reviewed	Question links	Exact matches	Direct matches	Partial matches	Mapped marks

Lecture structure and exam value

Slides	Lecture block	Historical signals	Exam value
2-4	Problem framing: man-made environment, cost, accept/reject/mitigate, designer task	2018 Q1(b); conceptual support to broader designer questions	Important bridge. Slide 2 directly matches the 2018 mitigation question; Slides 3-4 are prompts rather than full answer content.
5	Designer responsibilities - external + internal	2017 Q6(c); 2021 Q5(c); 2022 Q5(a); 2024 Q6(b)	Core high-return slide. Learn external wind/sun/precipitation and internal oxygen/temperature/humidity responsibilities.
6	Essential knowledge	2017 Q6(b); 2024 Q6(b)	Exact six-domain list: climate science, physiology, comfort, materials, passive technology, active technology.
7	Complexities	2017 Q6(c); 2020 Q5(a)	Direct repeated long-answer support: extremes, comfort vs openings, metabolic heat, energy/cost, composite climate, materials.
8	Reference	No direct question	Navigation only.

Priority note

If time is limited, memorize Slides 5-7 first. They form a compact 10-20 mark answer package. Slide 2 is a separate conceptual question and should be learned as the “mitigation” position.

2. Detailed Year-wise Question Mapping

Original paper references are preserved. Match labels mean: Exact = the slide block substantially answers the task; Direct = the requested concept is clearly taught but some expansion/sketch/application may still be needed; Partial = the deck only supports part of the requested answer.

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2016	Q5(a)	15	1	Write short notes on Context, Comfort and Man-made Environment.	Partial	2	The deck asks why human abode is a man-made environment but does not define all three terms. Use Lecture 01/02 for the complete answer.
2017	Q6(b)	10	2	Explain the essential knowledge for Climate Design.	Exact	6	Slide 6 gives the complete six-domain list expected by the course.
2017	Q6(c)	20	2	What are the responsibilities of the built-form designers? What kind of complexities do they have to face in Climate Design?	Exact	5-7	Slides 5 and 7 directly provide responsibilities and complexities; Slide 6 can strengthen the answer.
2018	Q1(b)	15	3	Can you defend your position about Climate and Built Environment: accept the fate, reject the development, or mitigate?	Exact	2	Slide 2 presents this exact decision frame and identifies mitigation as the course position.
2018	Q8(a)	10	4	Distinguish between passive and active means of thermal control. Which do you prefer?	Partial	6	Slide 6 names passive and active design technology only; it does not define or compare them.

2. Detailed Year-wise Question Mapping - Continued

Year	Original	Marks	QB p.	Question / task	Match	Lecture slides	Coverage assessment
2020	Q4(a)	15	5	Differentiate between active and passive thermal control techniques. Explain which one you prefer as a BECM professional.	Partial	6	The lecture confirms both as required knowledge but gives no comparison or preference logic.
2020	Q5(a)	15	5	What are the responsibilities of the built-form designers? What kinds of complexities do they have to deal with in climate design?	Exact	5-7	Direct repeat of the Lecture 04 core content.
2021	Q5(c)	10	6	Discuss about the designers' responsibilities in ensuring thermal comfort in built environment.	Direct	5	Internal temperature/humidity/oxygen plus external wind/sun/precipitation directly support the answer; thermal-comfort expansion may be added from Lecture 03/05.
2022	Q5(a)	10	8	Discuss the designer's responsibilities in ensuring thermal comfort in built environment. Mention the design tools for moderating micro climate.	Partial	5-6	Responsibilities are direct; the separate request for microclimate design tools belongs to later design lectures.
2024	Q6(b)	10	14	Discuss the designer's responsibilities in ensuring thermal comfort in built environment, and indicate the domains of essential knowledge a designer should have in this regard.	Exact	5-6	Slides 5 and 6 align almost one-to-one with the two requested parts.

3. Repeated Question Patterns and Quick Recognition

These patterns compress historical wording into the question families that matter for revision.

P	Normalized repeated question	Years seen	Slides	Why it matters
1	Designer responsibilities for climate/thermal comfort	2017, 2020, 2021, 2022, 2024	5	Most repeated family. External + internal responsibilities are the core answer skeleton.
2	Complexities faced by built-form designers in climate design	2017, 2020	7	Often paired with responsibilities; easy marks if memorized as a six-point list.
3	Essential knowledge for Climate Design	2017, 2024	6	Short, exact six-domain list. Recent 2024 paper revived it directly.
4	Accept fate / reject development / mitigate	2018	2	Unique wording, but conceptually central to the lecture and easy to recognize.
5	Passive vs active thermal control	2018, 2020	6 only	Related but not fully taught here; use another lecture/source for full comparison.

Quick recognition rules

Question clue	Go first to	What Lecture 04 can do
"responsibilities of built-form designer"	Slide 5	Exact external/internal list; add design examples if marks are high.
"essential knowledge" / "domains of knowledge"	Slide 6	Exact six-domain answer.
"complexities in climate design"	Slide 7	Exact list of six major complexities.
"accept fate / reject development / mitigate"	Slide 2	Exact conceptual frame; defend mitigation.
"passive vs active control"	Slide 6, then another source	Only names the two technologies; not enough for comparison.
"design tools for microclimate"	Do not stop at Lecture 04	Use later microclimate/design-specification lectures.

4. Slide-to-Question Reverse Index

Use this page when studying directly from the lecture deck: open the slide block, then practice the linked question family immediately.

Slides	Topic	Past questions	Study note
2	Why human abode is man-made environment; cost; accept/reject/mitigate	2016 Q5(a) partial; 2018 Q1(b)	Memorize mitigation as the preferred climate-responsive position.
3-4	Contemporary movement? What should designer do?	Supports 2017/2020/2021/2022/2024 responsibility questions	Prompt slides only; use them as answer introduction, not the full body.
5	Designer responsibilities	2017 Q6(c); 2020 Q5(a); 2021 Q5(c); 2022 Q5(a); 2024 Q6(b)	Highest-value slide. Build a two-column External vs Internal answer.
6	Essential knowledge	2017 Q6(b); 2024 Q6(b); partial support 2018/2020 passive-active	Learn the six items in order; do not pretend the slide compares passive vs active.
7	Complexities	2017 Q6(c); 2020 Q5(a)	Best written as a six-row table: issue + why difficult.
8	Reference	No direct question	No study time needed.

5. Coverage Gaps and Diagnostic Exclusions

Because Lecture 04 is short, it is especially important not to over-credit it. It answers responsibilities, knowledge and complexities very well, but several adjacent exam questions require later technical lectures.

Gap	Affected questions	Action
Full passive vs active thermal-control definitions, comparison, examples and preference	2018 Q8(a); 2020 Q4(a)	Lecture 04 only lists passive/active technology under essential knowledge. Use the dedicated thermal-control material.
Design tools for moderating microclimate	2022 Q5(a)	Responsibilities are here; tools such as layout, spacing, shading and envelope response are in later design lectures.
Complete definitions of climate, context, comfort and man-made environment	2016 Q5(a) and related 2017/2018/2024 conceptual questions	Use Lectures 01-02. Lecture 04 merely recalls the man-made-environment problem.
Detailed mechanisms of thermal comfort / human heat exchange	2021 Q5(c), 2024 Q6(b) if expanded	Use Lectures 03 and 05 for physiology and comfort indices.

Mapping discipline	A question is marked Exact only when the deck itself supplies the requested content. The presence of the words “passive” and “active” on Slide 6 is not treated as a full passive-vs-active answer.
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6. Recommended Study Plan

This lecture is best studied as three lists plus one concept. Because the content is compact, a full exam-ready revision can be completed quickly.

Step	Study block	Slides	Practice immediately
1	Designer responsibilities - External vs Internal	5	2017 Q6(c); 2020 Q5(a); 2021 Q5(c); 2022 Q5(a); 2024 Q6(b)
2	Essential knowledge - six domains	6	2017 Q6(b); 2024 Q6(b)
3	Complexities - six issues	7	2017 Q6(c); 2020 Q5(a)
4	Mitigation position	2	2018 Q1(b)
5	Gap handoff: passive/active and microclimate tools	6 + later lectures	2018 Q8(a); 2020 Q4(a); 2022 Q5(a)

Exam-ready minimum from this lecture

Priority	What to reproduce	Exam technique
A	Slide 5 responsibilities	Two-column External/Internal table; add one design implication per item.
A	Slide 6 essential knowledge	Six-item list with one-line purpose for each.
A	Slide 7 complexities	Six-row issue/why-difficult table.
B	Slide 2 mitigation	State why rejection is unrealistic and mitigation is climate-responsive.

Bottom line

Slides 5-7 are the real scoring core. If you can reproduce those three slides accurately and connect them to climate-responsive design, Lecture 04 becomes a reliable 10-20 mark support block.